ECET 250E

Linear Circuits 1

Week 4 | Chapter 4: Circuit Theorems

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Electronics & Computer Engineering Technology

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Based on Fundamentals of Electric Circuits, 2026 Release — C.K. Alexander, M.N.O. Sadiku & S.J. Fredette (McGraw Hill)

Overview

- In this chapter, the concept of **superposition** will be introduced.
- Source transformation will also be covered.
- **Thevenin and Norton's** theorems will be covered.
- And Maximum Power Transfer.

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Linearity

- Linearity in a circuit means that as current is changed, the voltage changes proportionally.
- It also requires that the response of a circuit to a sum of sources will be the sum of the individual responses from each source separately.
- A resistor satisfies both of these criteria.

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Superposition

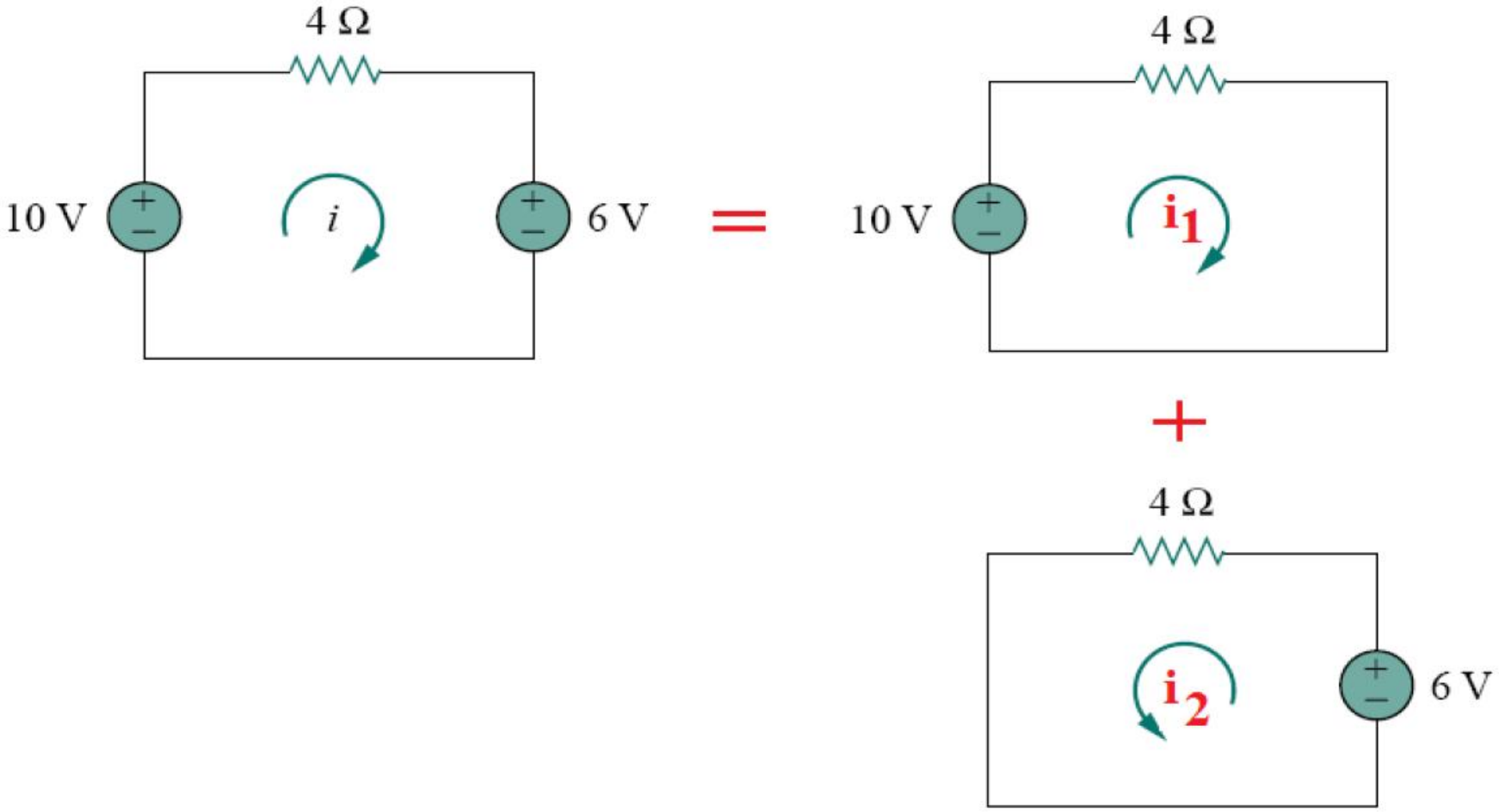
If there are two or more independent sources there are two ways to solve for the circuit parameters:

- Nodal or mesh analysis.
- Use superposition.

The superposition principle states that the voltage across (or current through) an element in a linear circuit is the algebraic sum of the voltages across (or currents through) that element due to each independent source acting alone.

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If a circuit has two or more independent sources, one way to determine the value of a specific variable (voltage or current) is to use nodal or mesh analysis as in Chapter 3. Another way is to determine the contribution of each independent source to the variable and then add them up. The latter approach is known as the *superposition*.



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Applying Superposition

- Using superposition means applying one independent source at a time.
- Dependent sources are left alone.
- The steps are:

1. Turn off all independent sources except one source. Find the output (voltage or current) due to that active source using the techniques covered in Chapters 2 and 3.
2. Repeat step 1 for each of the other independent sources.
3. Find the total contribution by adding algebraically all the contributions due to the independent sources.

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To apply the superposition principle, we must keep two things in mind:

1. We consider one independent source at a time while all other independent sources are *turned off* **(set to zero)**

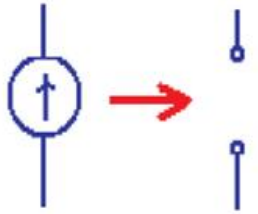
   This implies that we

   replace every voltage source by a short circuit,

   and

   every current source by an open circuit.

2. Dependent sources are left intact because they are controlled by circuit variables.

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Example 4.3

Use the superposition theorem to find v in the circuit of Fig. 4.6.

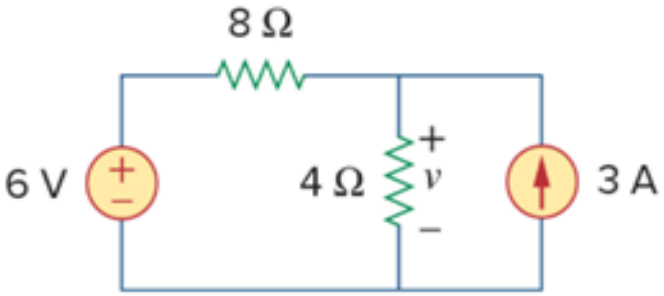


Figure 4.6
For Example 4.3.

Practice Problem 4.3

Using the superposition theorem, find v_o in the circuit of Fig. 4.8.

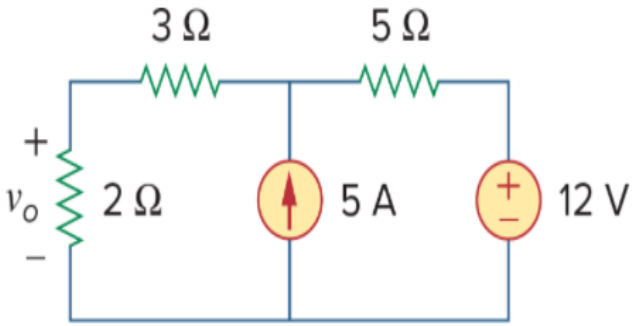


Figure 4.8
For Practice Prob. 4.3.

Answer: 7.4 V.

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Source Transformation

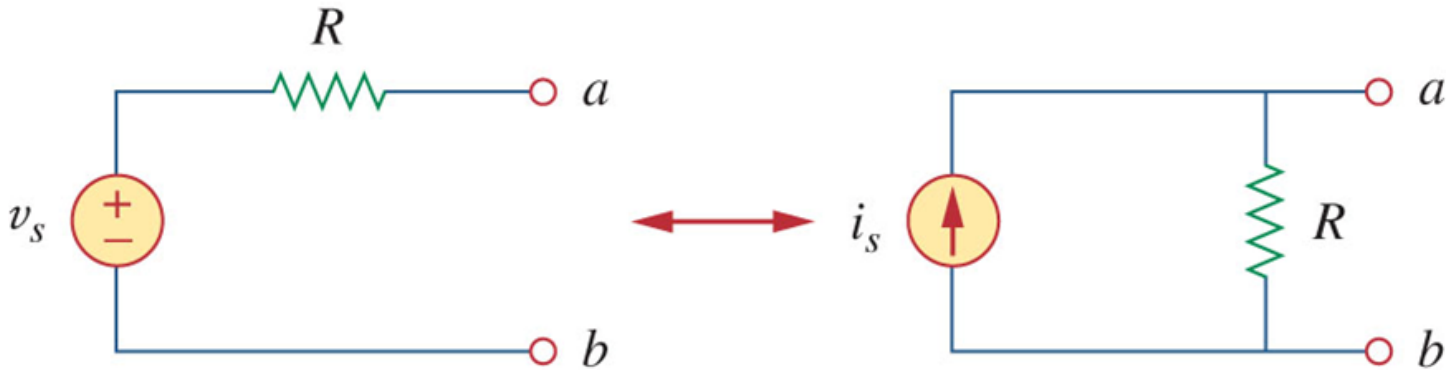
- it is possible to transform a source from one form to another.
- This can be useful for simplifying circuits.
- The principle behind all of these transformations is equivalence.

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Source Transformation II

- A source transformation is the process of replacing a voltage source v_s in series with a resistor R by a current source i_s in parallel with a resistor R, or vice versa.

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Terminal Equivalency

- These transformations work because the two sources have equivalent behavior at their terminals.
- If the sources are turned off the resistance at the terminals are both R.
- If the terminals are short circuited, the currents need to be the same.
- From this we get the following requirement:

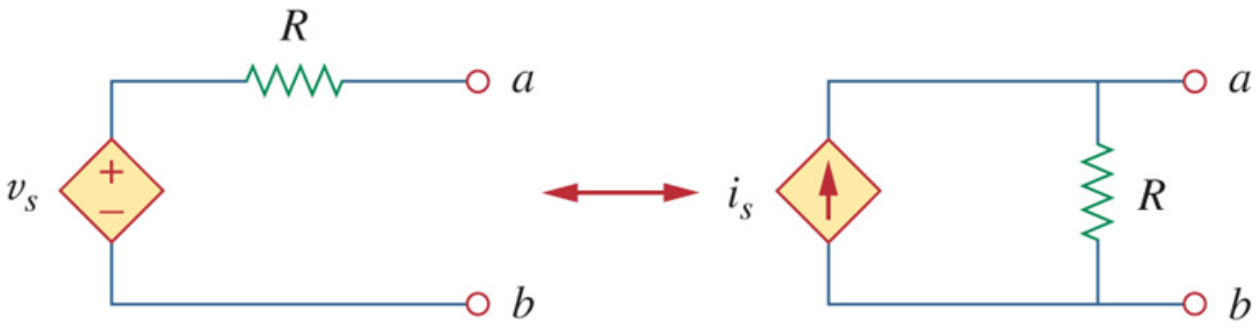
$$v_s = i_s R \quad \text{or} \quad i_s = \frac{v_s}{R}$$

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Dependent Sources

- Source transformation also applies to dependent sources.
- But, the dependent variable must be handled carefully.
- The same relationship between the voltage and current holds here:

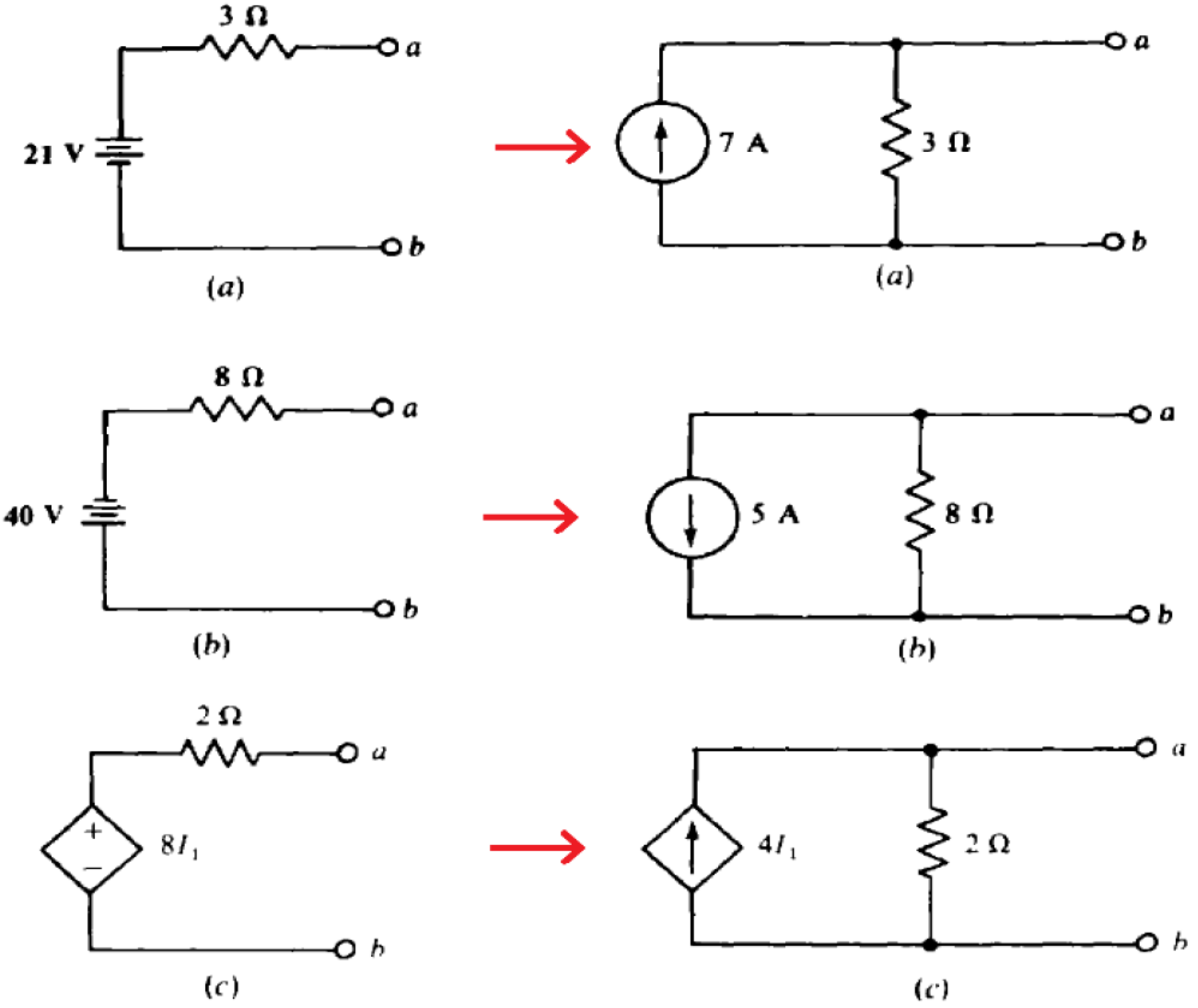
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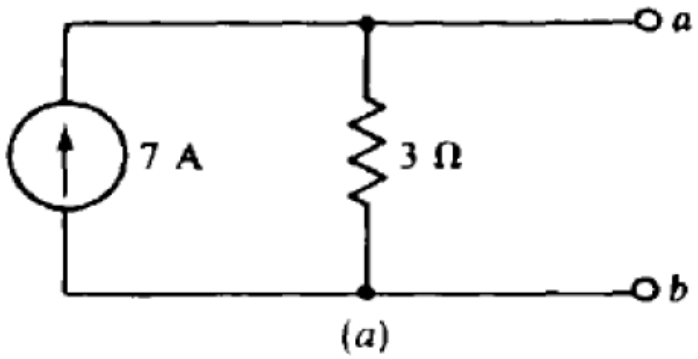


Source transformation rules

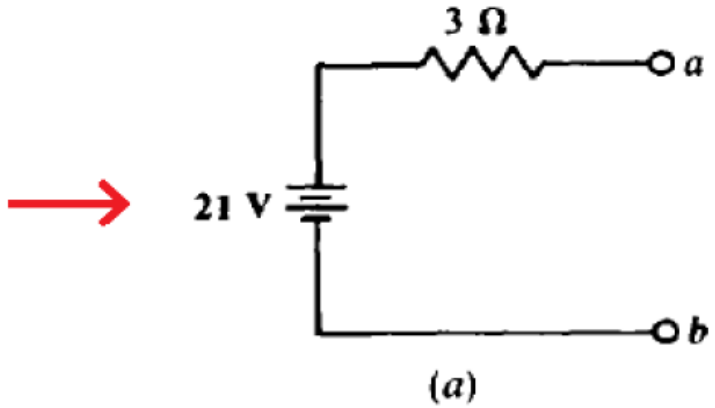
- Note that the arrow of the current source is directed towards the positive terminal of the voltage source.
- Source transformation is not possible when $R = 0$ for an ideal voltage source.
- For a realistic source, $R \neq 0$.
- For an ideal current source, $R = \infty$ also prevents the use of source transformation.

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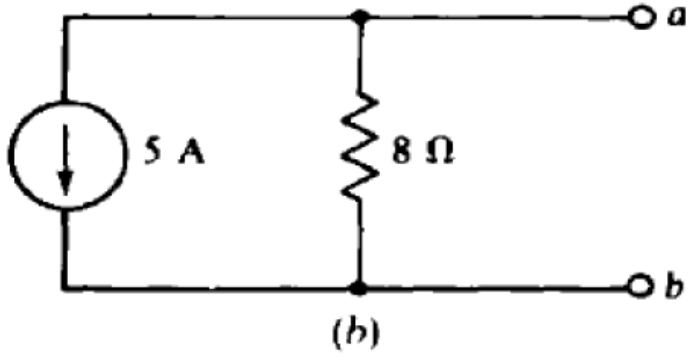

3 Ω
a
21 V
b
(a)
7 A
3 Ω
a
b
(a)
8 Ω
a
40 V
b
(b)
5 A
8 Ω
a
b
(b)
2 Ω
a
+
−
8I1
b
(c)
4I1
2 Ω
a
b
(c)



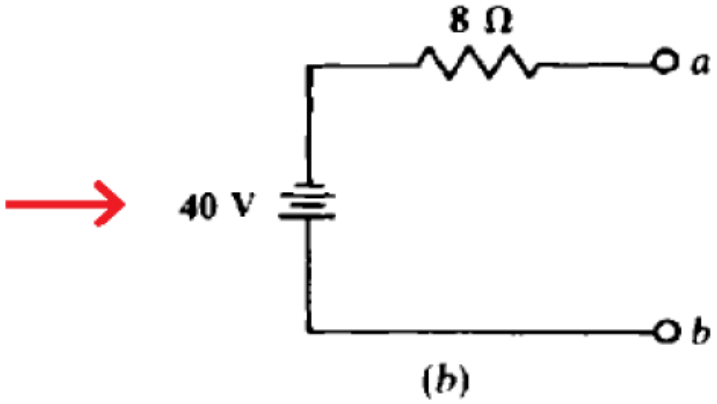
(a)



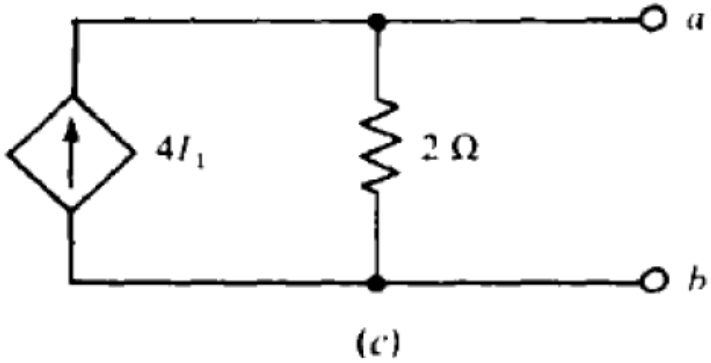
(a)



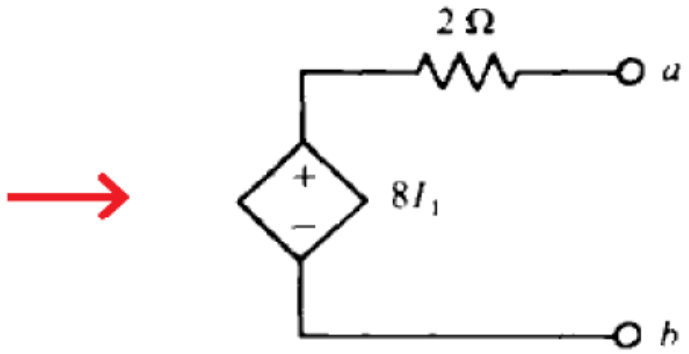
(b)



(b)



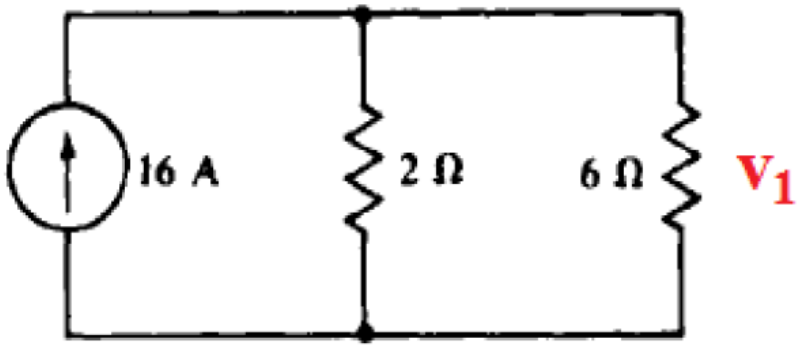
(c)



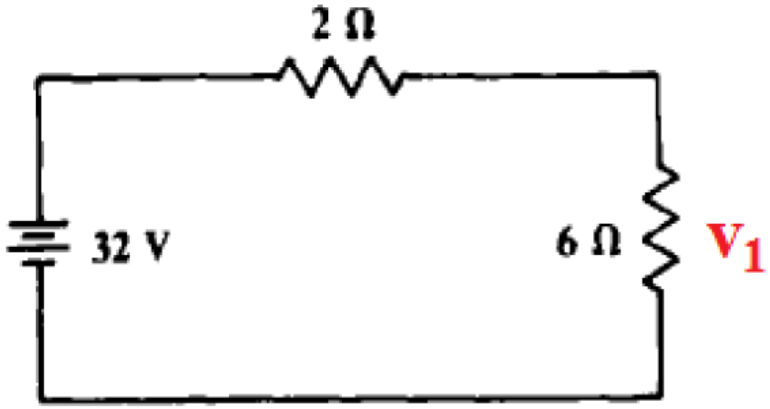
(c)

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We are interested to find v1, which circuit looks easier to solve ?



(a)



(b)

Circuit (a) can be transformed as circuit (b)

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NOTE

- Source transform is done only in one or two parts (few part) of the circuit containing voltage or current sources.
- We don’t need to transform the whole circuit

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Example 4.6

Use source transformation to find v_o in the circuit of Fig. 4.17.

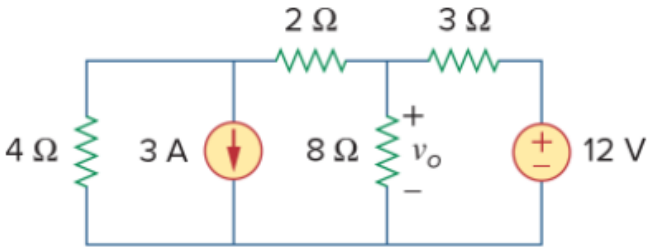


Figure 4.17
For Example 4.6.

Example 4.7

Find v_x in Fig. 4.20 using source transformation.

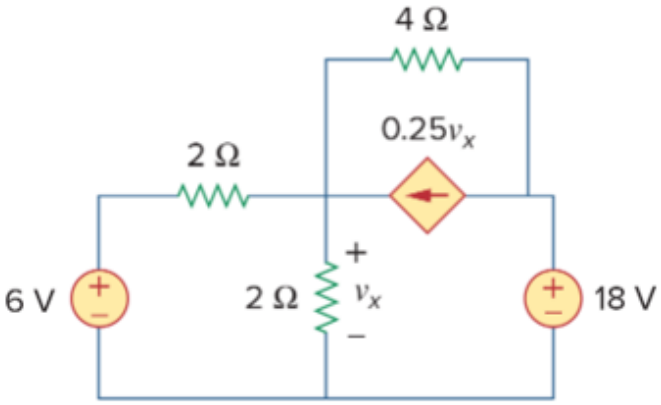


Figure 4.20
For Example 4.7.

Thevenin's Theorem

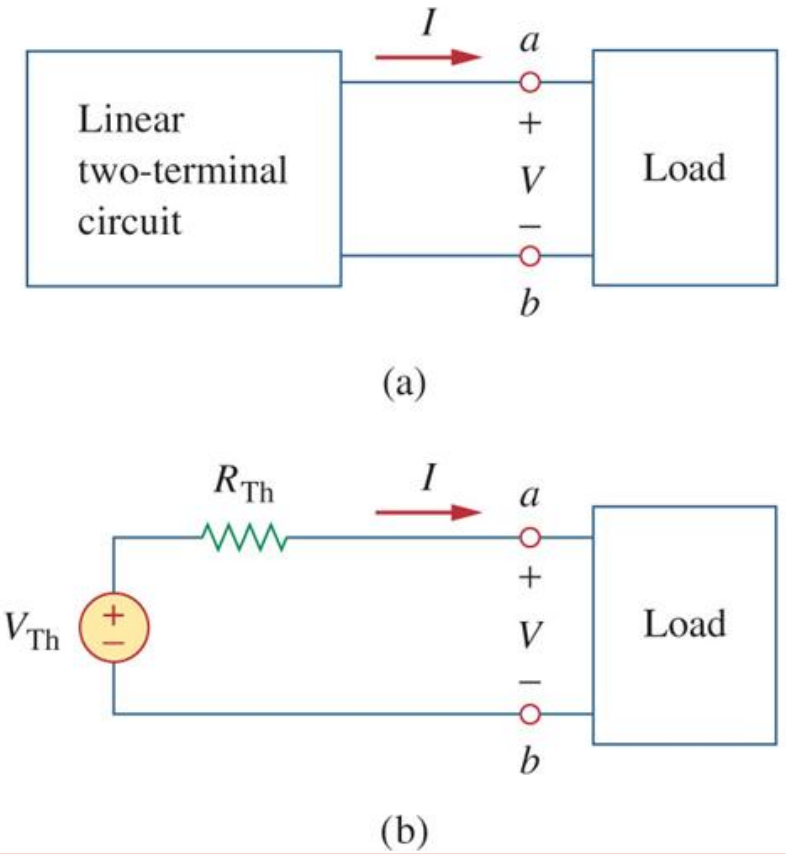
- In many circuits, one element will be variable.
- An example of this is mains power; many different appliances may be plugged into the outlet, each presenting a different resistance.
- This variable element is called the load.
- Ordinarily one would have to reanalyze the circuit for each change in the load.

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Thevenin's Theorem II

- Thevenin's theorem states that a linear two terminal circuit may be replaced with a voltage source and resistor.
- The voltage source's value is equal to the open circuit voltage at the terminals.
- The resistance is equal to the resistance measured at the terminals when the independent sources are turned off.

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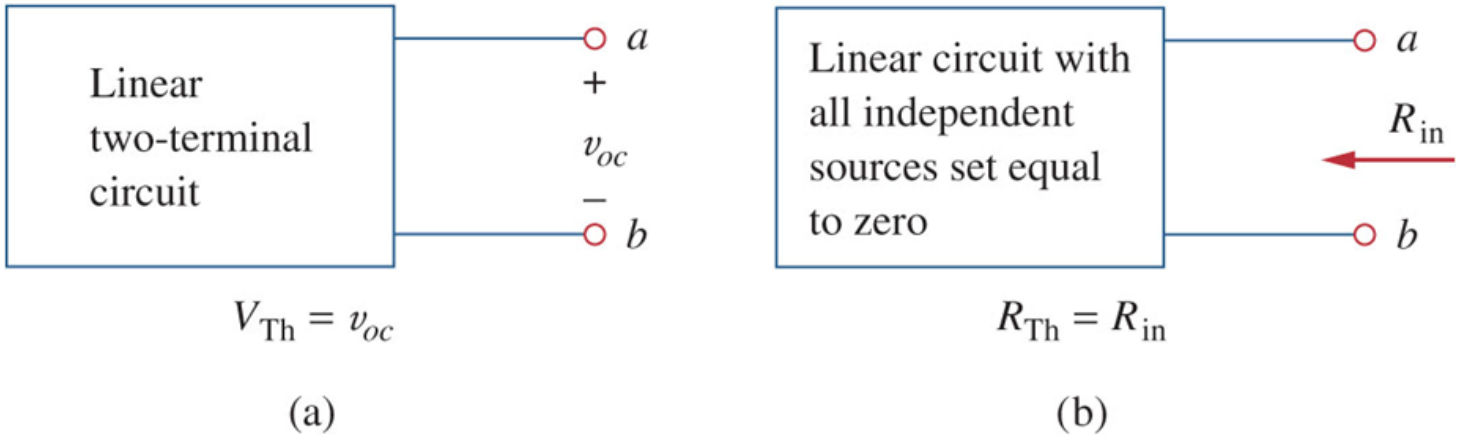


Thevenin's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a voltage source V_{Th} in series with a resistor R_{Th}, where V_{Th} is the open-circuit voltage at the terminals and R_{Th} is the input or equivalent resistance at the terminals when the independent sources are turned off.

Thevenin's Theorem III

- There are two cases to consider when finding the equivalent resistance.
- Case 1: If there are no dependent sources, then the resistance may be found by simply turning off all the sources.

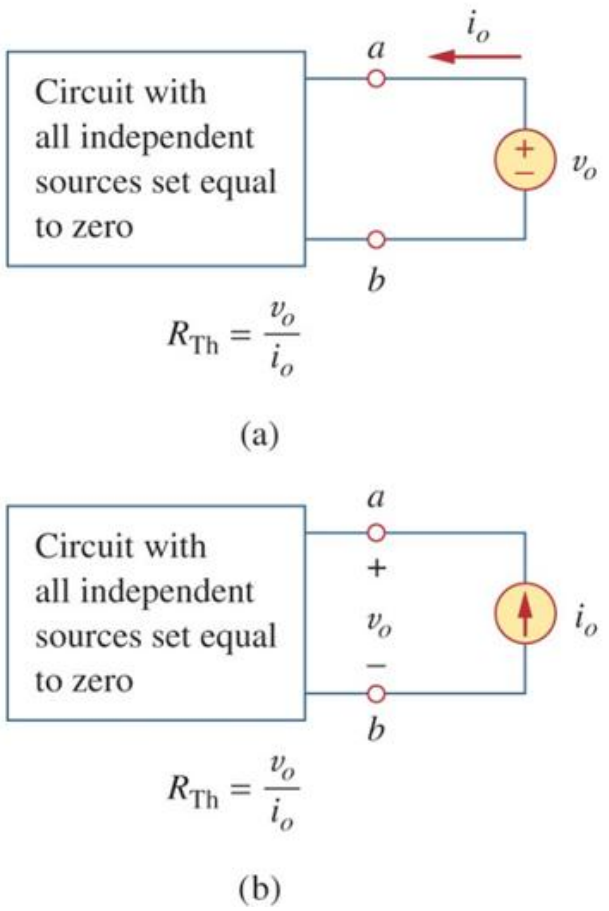
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Thevenin's Theorem IV

- Case 2: If there are dependent sources, we still turn off all the independent sources.
- Now apply a voltage v_0 (or current i_0) to the terminals and determine the current i_0 (voltage v_0).

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Thevenin's Theorem V

- Thevenin's theorem is very powerful in circuit analysis.
- It allows one to simplify a circuit.
- A large circuit may be replaced by a single independent voltage source and a single resistor.
- The equivalent circuit behaves externally exactly the same as the original circuit.

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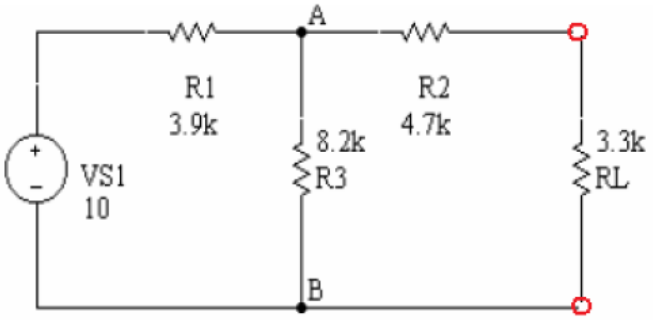
Negative Resistance?

- It is possible for the result of this analysis to end up with a negative resistance.
- This implies the circuit is supplying power.
- This is reasonable with dependent sources.
- Note that in the end, the Thevenin equivalent makes working with variable loads much easier.
- Load current can be calculated with a voltage source and two series resistors.
- Load voltages use the voltage divider rule.

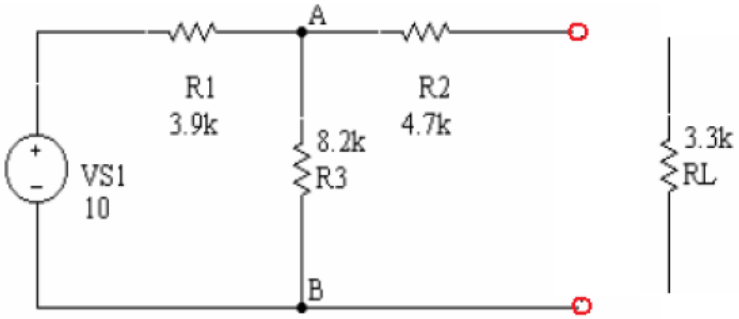
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Steps for Thevenin's Equivalent

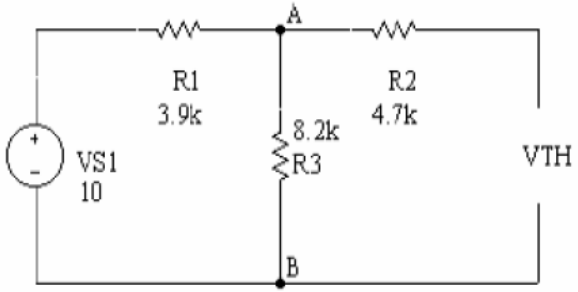
1. **Identify the Load resistance**



2. **Separate the load resistance**



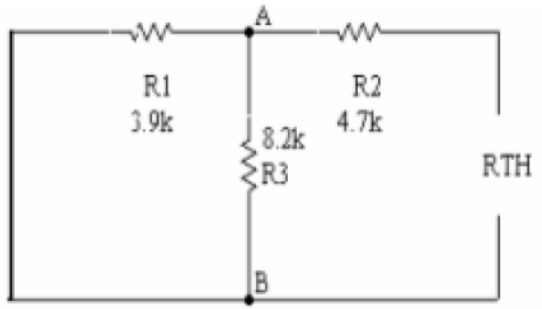
3. **Calculate the open circuit voltage V_{TH}**



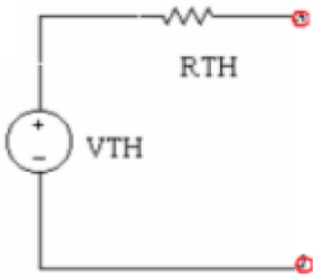
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Steps for Thevenin's Equivalent ….

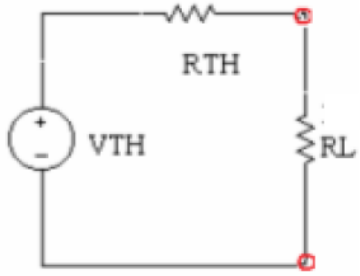
4. Replace all current sources with an open
   Replace voltage sources with a short
   Calculate R_{TH}



5. Draw Thevenin equivalent circuit by putting value of V_{TH} and R_{TH}



6. Connect the load R_L

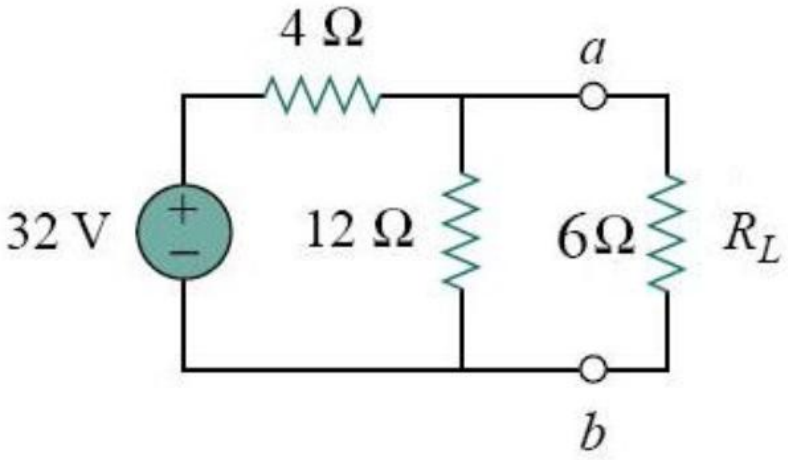


Now we can easily calculate the load current & voltage etc.

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EXAMPLE

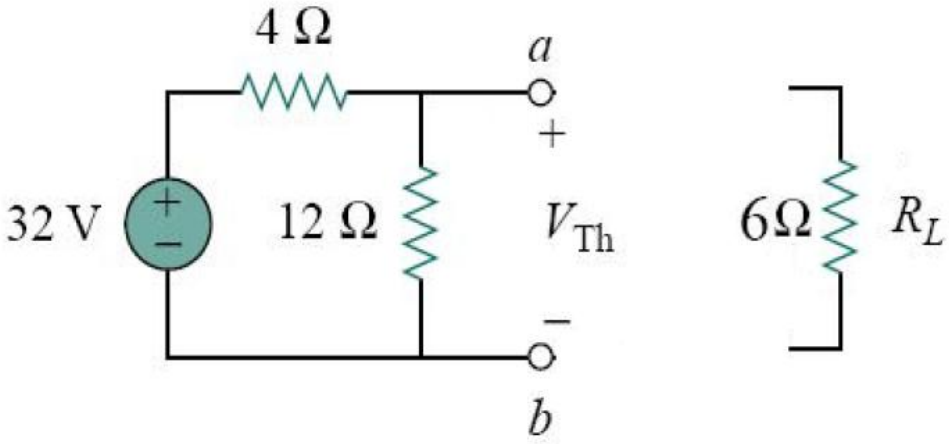
Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a-b. Then find the current through R_L



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Solution:

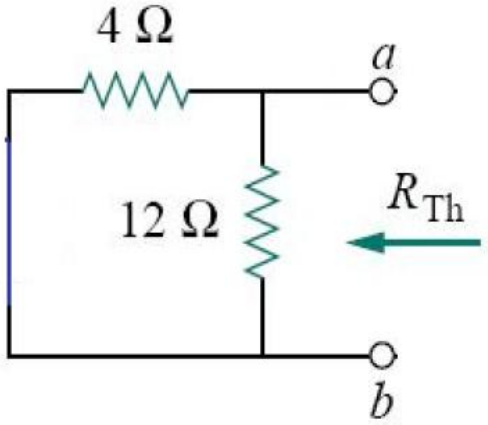
We separate the load



Calculate V_{Th}

$$V_{Th} = V_{ab} = V_{12} = \frac{12}{4+12}(32) = 24\ V$$

Next, We find R_{Th} by turning off the 32-V voltage source (replacing it with a short circuit)

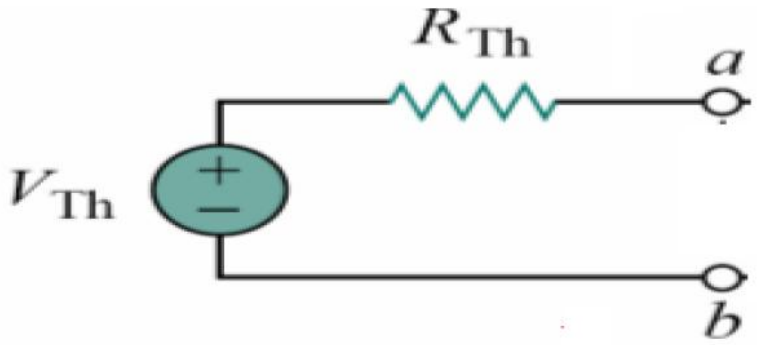


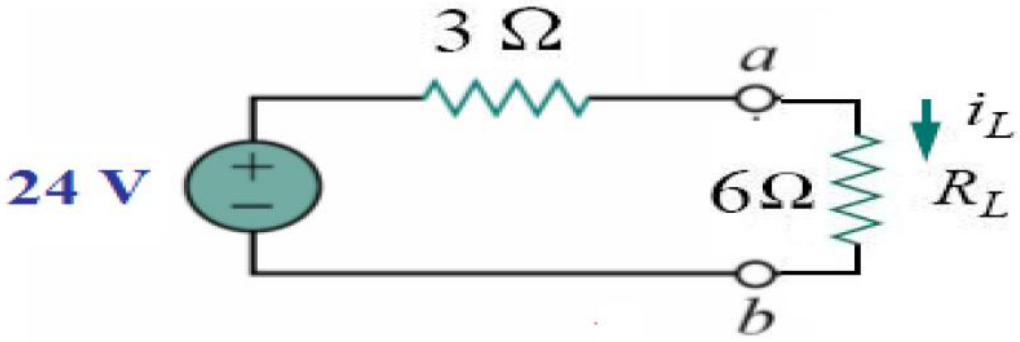
$$R_{Th} = 4 \parallel 12 = \frac{4 \times 12}{16} = 3\ \Omega$$

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$V_{Th} = 24\ V$

$R_{Th} = 3\ \Omega$



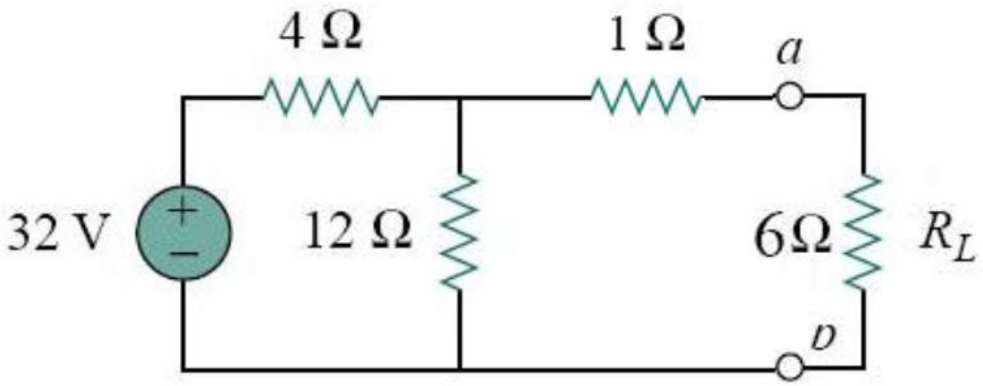


$$I_L = \frac{24}{3+6} = 2.67A$$

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Example:

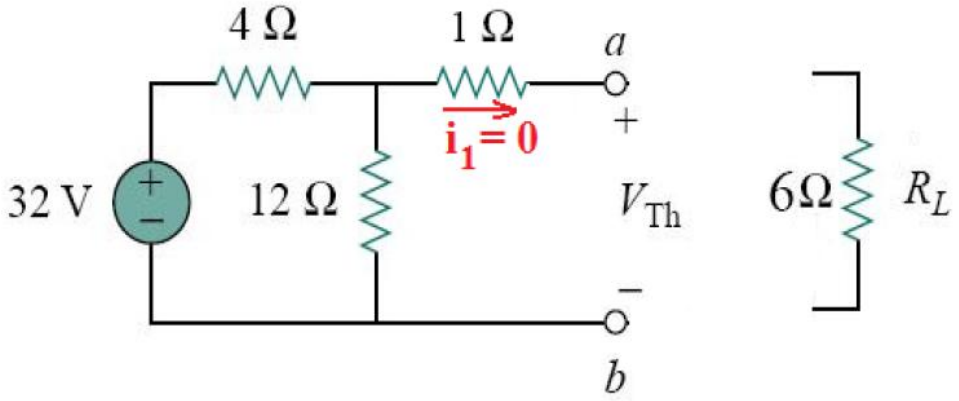
Find the Thevenin equivalent circuit of the circuit shown in Fig, to the left of the terminals a-b. Then find the current through R_L



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Solution:

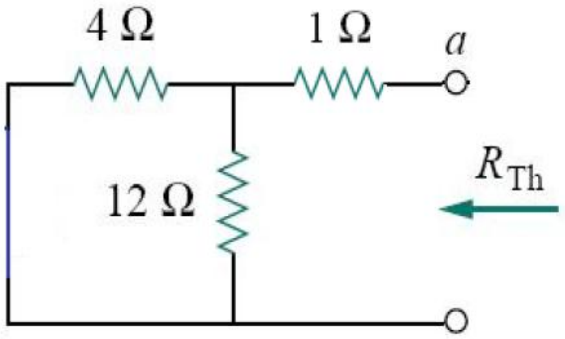
We separate the load



Calculate V_{Th}

$$V_{Th} = V_{ab} = V_{12} = \frac{12}{4+12}(32) = 24\ \text{V}$$

Next, We find R_{Th} by turning off the 32-V voltage source (replacing it with a short circuit)

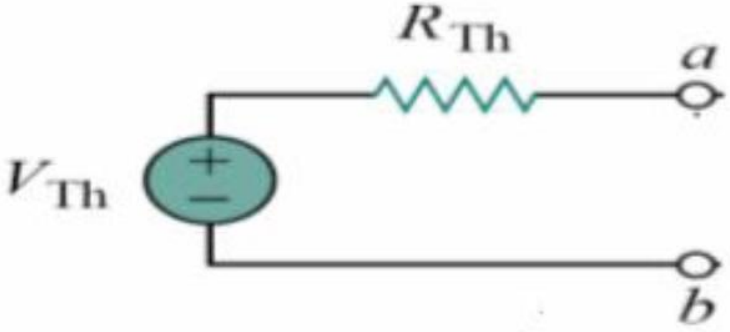


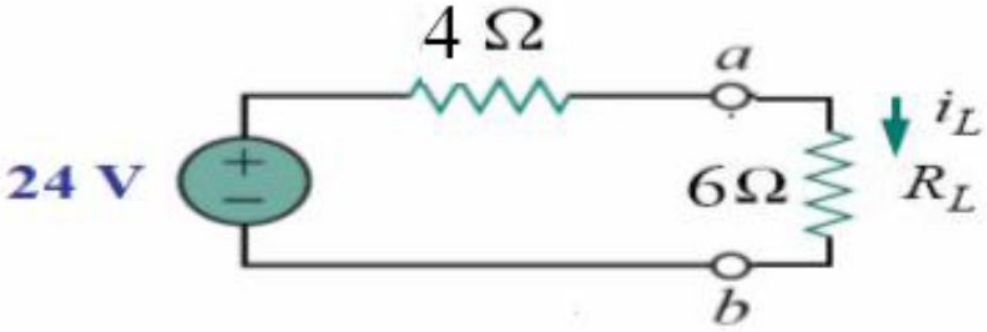
$$R_{Th} = 4 \parallel 12 + 1 = \frac{4 \times 12}{16} + 1 = 4\ \Omega$$

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$V_{Th} = 24\ V$

$R_{Th} = 4\ \Omega$





$$I_L = \frac{24}{4+6} = 2.4\ A$$

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Example 4.8

Find the Thevenin equivalent circuit of the circuit shown in Fig. 4.27, to the left of the terminals *a-b*. Then find the current through $R_L = 6 \quad \Omega$.

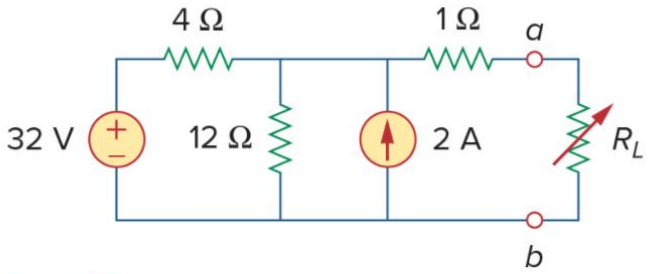


Figure 4.27
For Example 4.8.

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Practice Problem 4.8

Using Thevenin's theorem, find the equivalent circuit to the left of the terminals in the circuit of Fig. 4.30. Then find I.

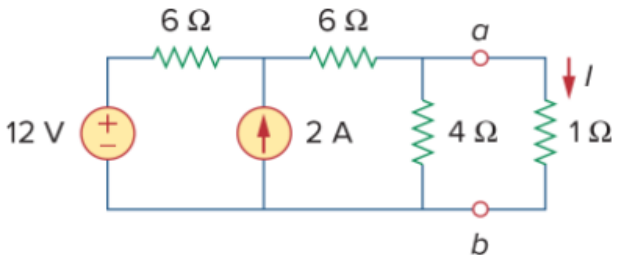


Figure 4.30
For Practice Prob. 4.8.

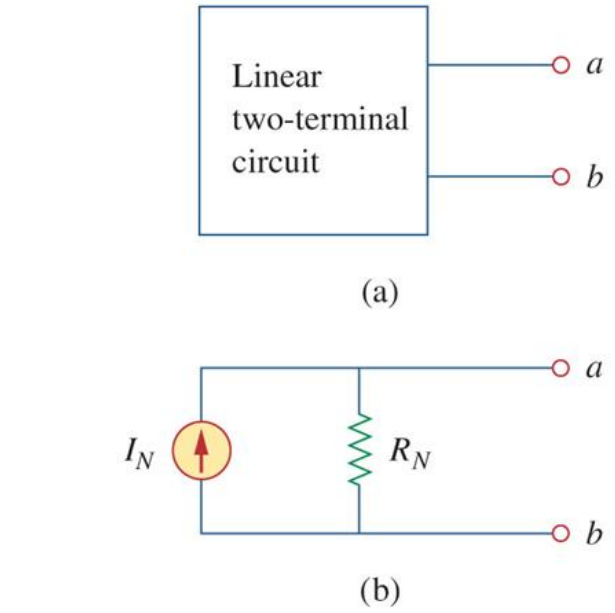
Answer: $V_{Th} = 6$ V, $R_{Th} = 3\ \Omega$, $I = 1.5$ A.

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Norton's Theorem

- Similar to Thevenin's theorem, Norton's theorem states that a linear two terminal circuit may be replaced with an equivalent circuit containing a resistor and a current source.
- The Norton resistance will be exactly the same as the Thevenin.

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Norton's theorem states that a linear two-terminal circuit can be replaced by an equivalent circuit consisting of a current source I_N in parallel with a resistor R_N, where I_N is the short-circuit current through the terminals and R_N is the input or equivalent resistance at the terminals when the independent sources are turned off.

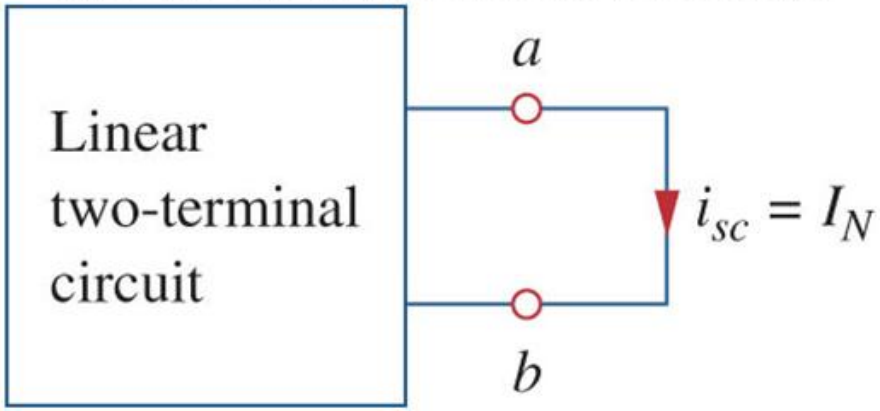
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Norton's Theorem II

- The Norton current I_N is found by short circuiting the circuit's terminals and measuring the resulting current.

$$I_N = i_{sc}$$

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Norton versus Thevenin

- These two equivalent circuits can be related to each other.
- One need only look at source transformation to understand this.
- The Norton current and Thevenin voltage are related to each other as follows:

$$I_N = \frac{V_{Th}}{R_{Th}}$$

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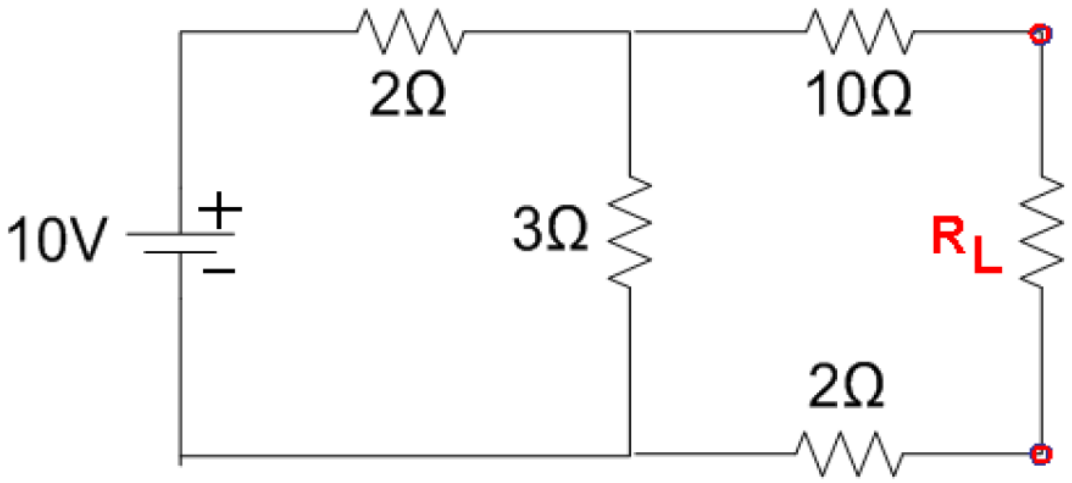
Norton versus Thevenin II

- With V_{TH}, I_N, and (R_{TH}=R_N) related, finding the Thevenin or Norton equivalent circuit requires that we find:
- The open-circuit voltage across terminals *a and b*.
- The short-circuit current at terminals *a and b*.
- The equivalent or input resistance at terminals *a and b when* all independent sources are turned off.

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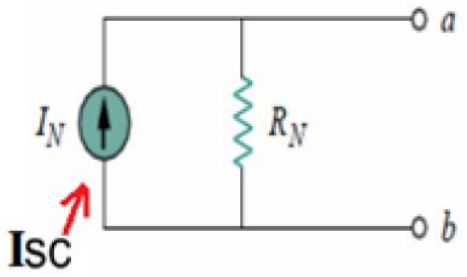
Example

Find Norton's equivalent circuit and current thru R_L, if $R_L = 1\Omega$

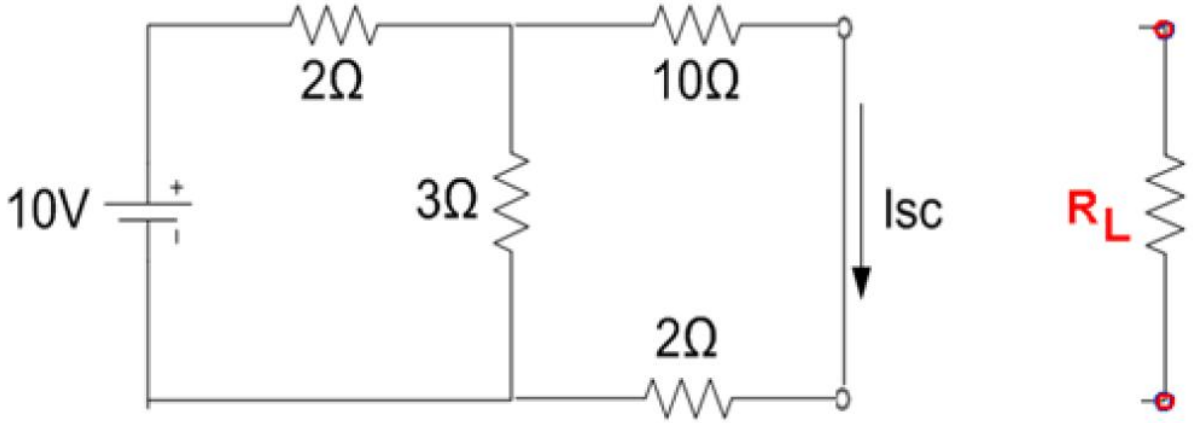


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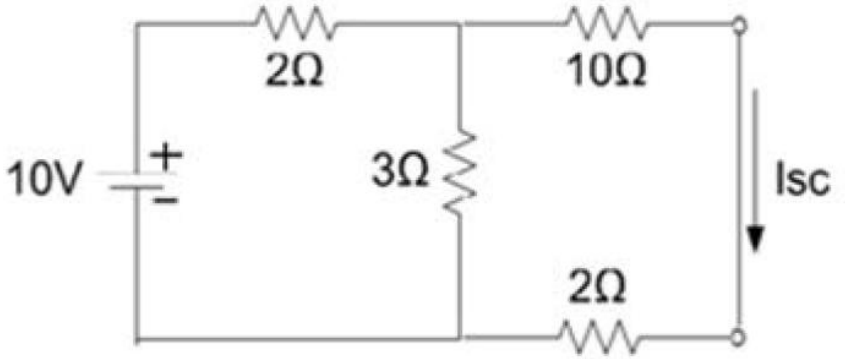
To find I_N (or I_{sc}), we remove the load resistance and short circuit the terminals.

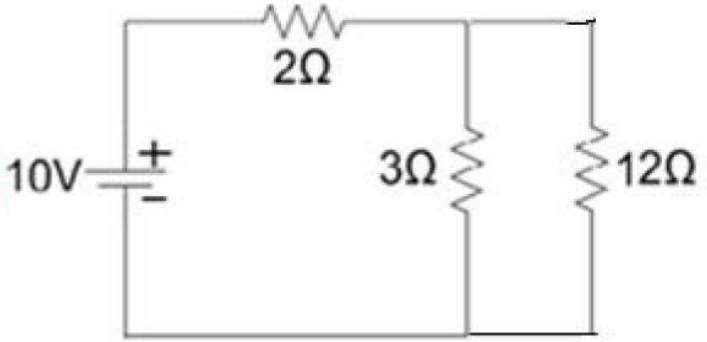


Find I_N



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Find R total

$$2+3\|(10+2)$$

$$=2+\frac{3\times12}{3+12}=4.4\Omega$$



Find I total

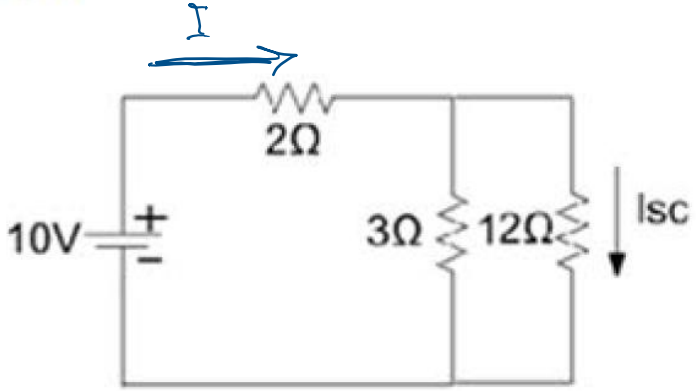
$$I=\frac{V}{R}=\frac{10}{4.4}=2.27A$$

Current divider

$$I_{SC}=\frac{3\|12}{12}\times2.27$$

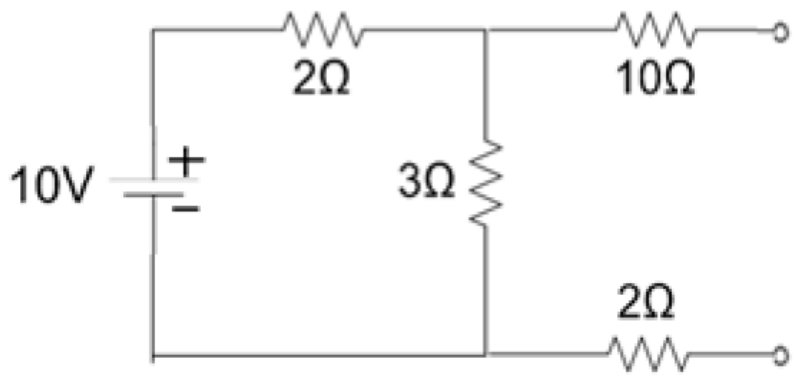
$$\frac{3}{12+3}\times2.27$$

$$=0.45A$$

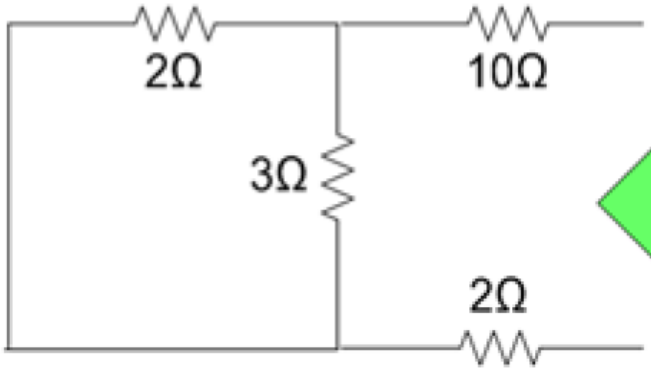


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R'_{TH} is found in the same way as in case of Thevenin theorem.



To find R_N (or R_{TH}), we short the voltage source.



R_N

$$R_N = 10 + 2 \parallel 3 + 2$$

$$= 10 + \frac{2 \times 3}{2 + 3} + 2$$

$$= 13.2\Omega$$

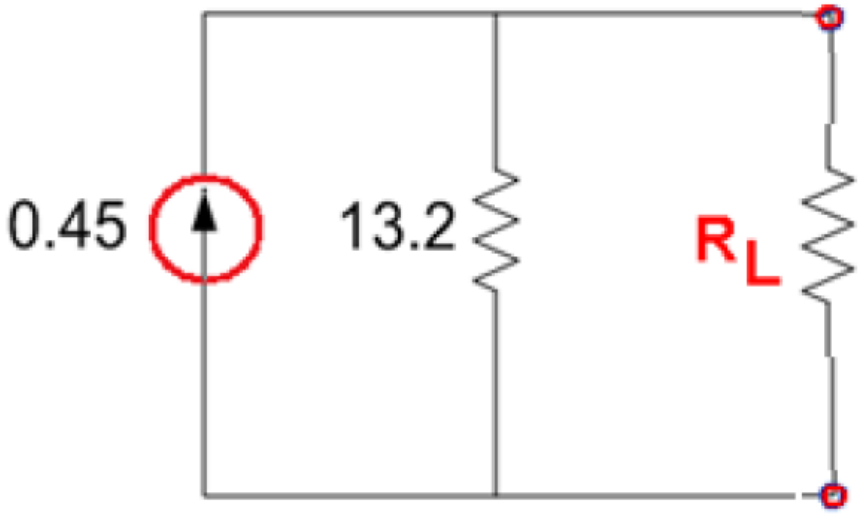
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We calculated,

$$I_N = I_{SC} = 0.45A$$

$$R_N = 13.2\Omega$$

Therefore, **Norton's equivalent circuit**



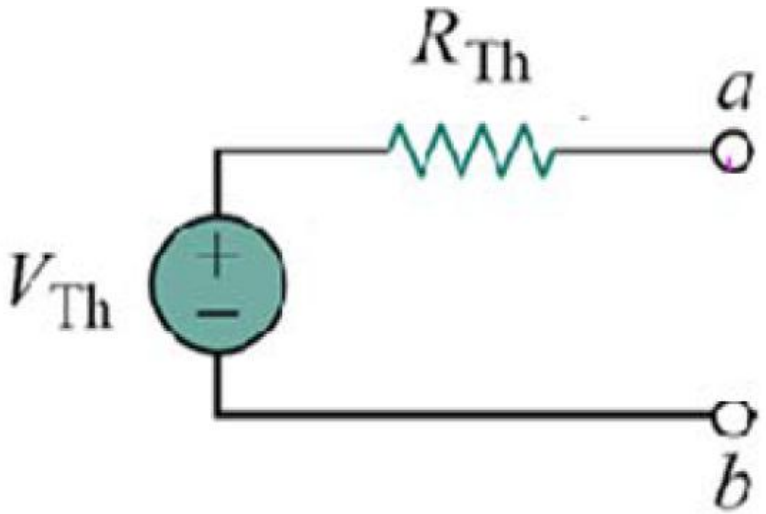
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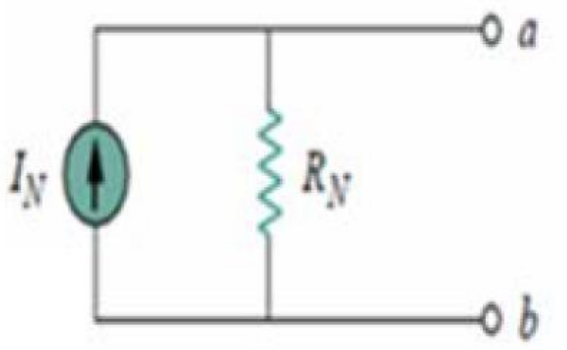
Relationship Between Thevenin's and Norton's Circuit

$$V_{\text{Th}} = v_{oc}$$

$$I_N = i_{sc}$$

$$R_{\text{Th}} = \frac{v_{oc}}{i_{sc}} = R_N$$





Example 4.11

Find the Norton equivalent circuit of the circuit in Fig. 4.39 at terminals *a-b*.

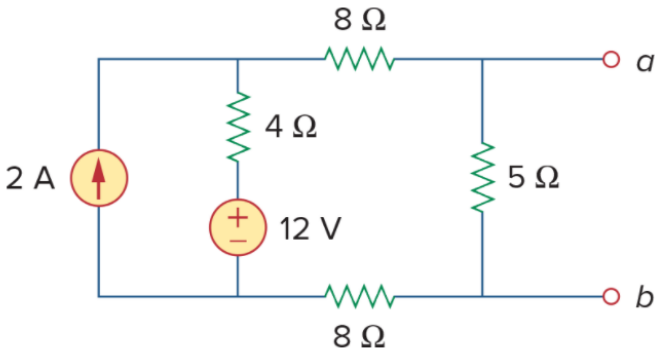


Figure 4.39
For Example 4.11.

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Equivalent Circuits with Dependent Sources

We cannot find R_{TH} in circuits with dependent sources using the total resistance method.

But we can use

$$R_{TH} = \frac{V_{OC}}{I_{SC}}$$

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Practice Problem 4.11

Find the Norton equivalent circuit for the circuit in Fig. 4.42, at terminals *a-b*.

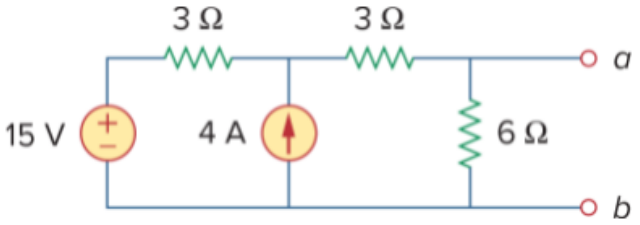


Figure 4.42
For Practice Prob. 4.11.

Answer: $R_N = 3\ \Omega, I_N = 4.5\ \text{A}$.

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Maximum Power Transfer

- In many applications, a circuit is designed to power a load.
- Among those applications there are many cases where we wish to maximize the power transferred to the load.
- Unlike an ideal source, internal resistance will restrict the conditions where maximum power is transferred.

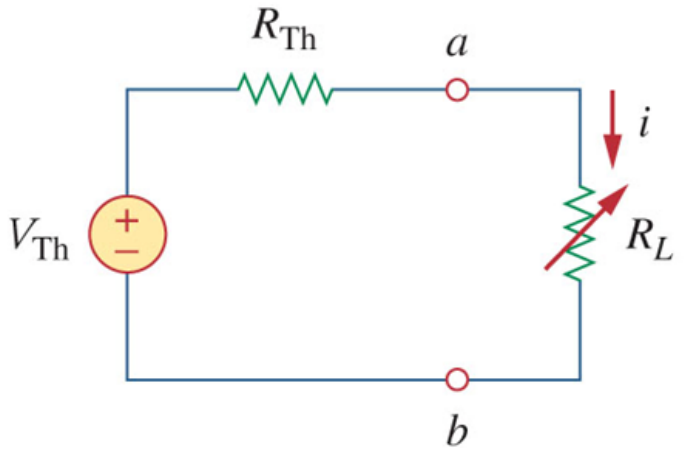
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Maximum Power Transfer II

- We can use the Thevenin equivalent circuit for finding the maximum power in a linear circuit.
- We will assume that the load resistance can be varied
- Looking at the equivalent circuit with load included, the power transferred is:

$$p = \left(\frac{V_{Th}}{R_{Th} + R_L} \right)^2 R_L$$

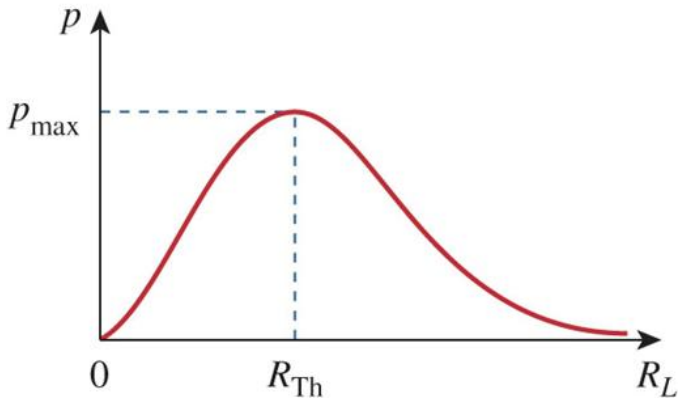
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Maximum Power Transfer III

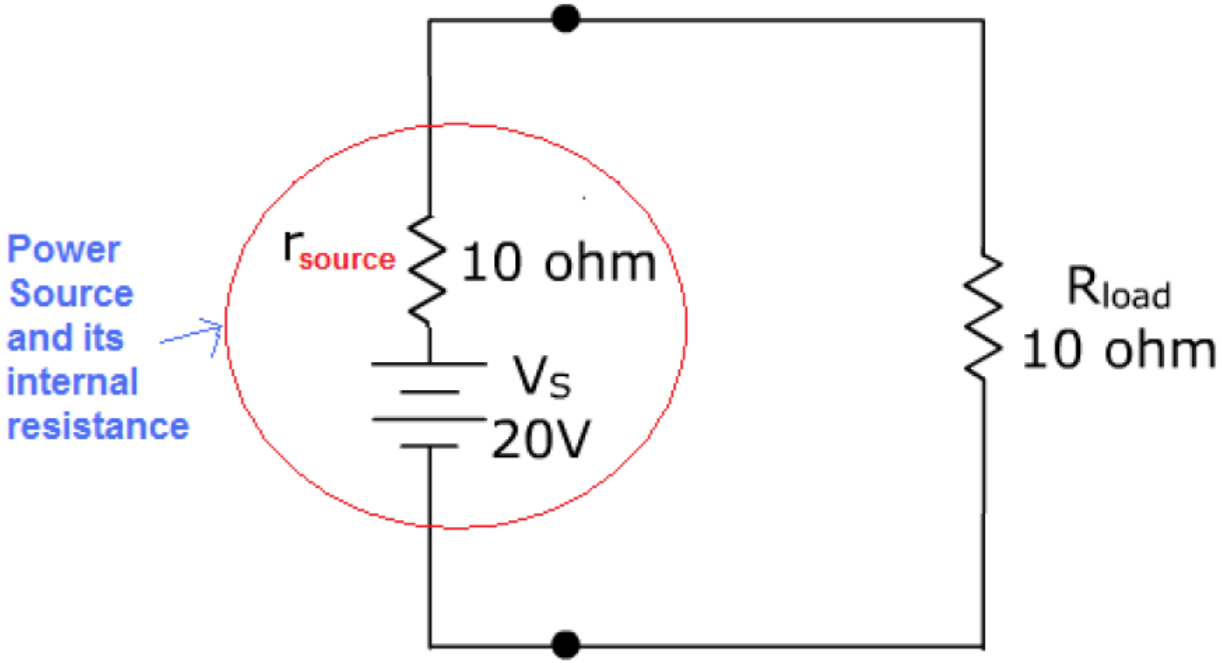
- For a given circuit, V_{TH} and R_{TH} are fixed. By varying the load resistance R_L, the power delivered to the load varies as shown.
- You can see that as R_L approaches 0 and ∞ the power transferred goes to zero.
- In fact the maximum power transferred is when $R_L = R_{TH}$.

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Maximum power is transferred from the source to the load when the resistance of the source is equal to the resistance of the load.

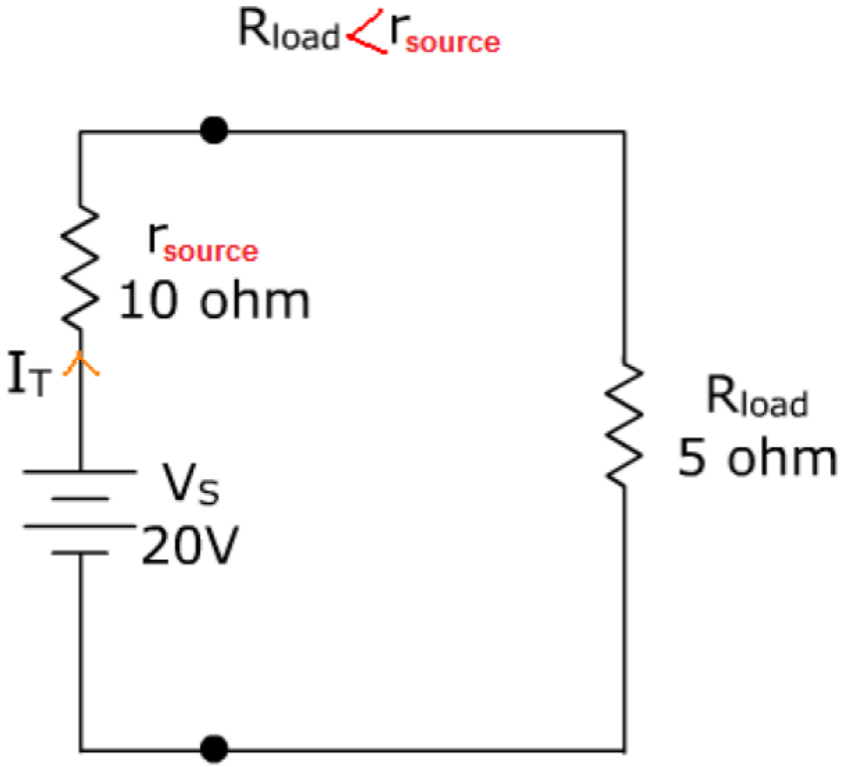
$$(R_{load} = r_{source})$$



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Let's look at some examples.

Example 1



$$R_T = r_{source} + R_{load}$$

$$R_T = 10\ \Omega + 5\ \Omega$$

$$R_T = 15\ \Omega$$

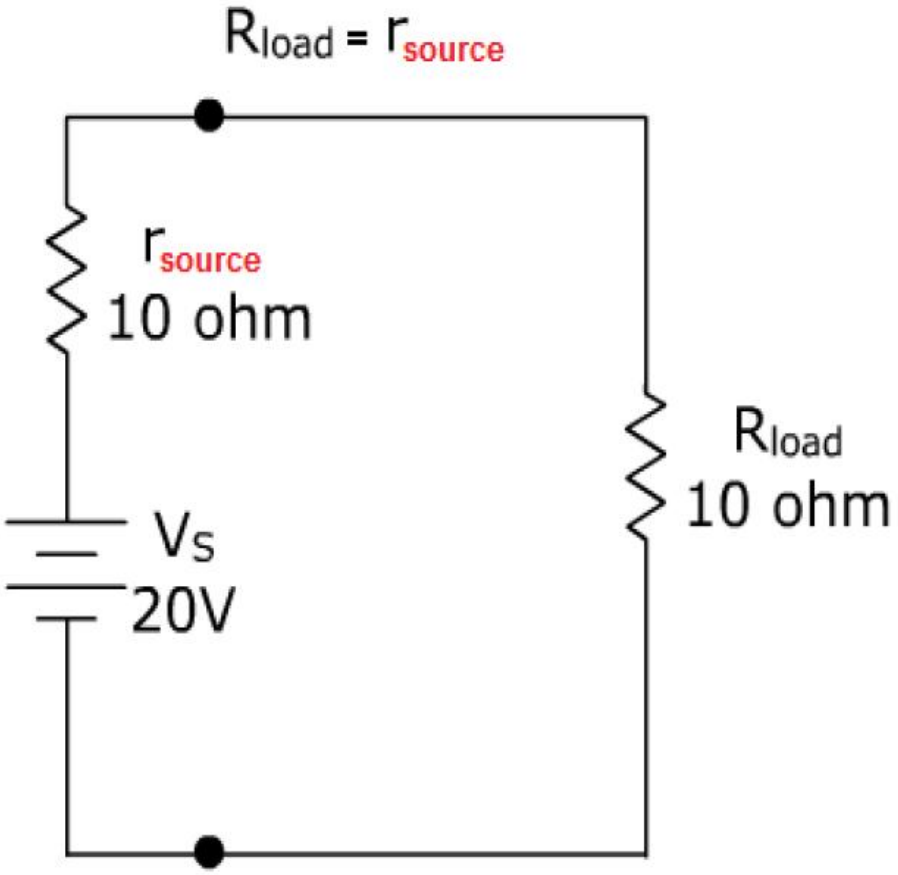
$$I_T = \frac{V_S}{R_T} = \frac{20V}{15\Omega} = 1.33\ A$$

$$P_L = I^2 R_{load} = (1.33\ A)^2\ (5\ \Omega)$$

$$\mathbf{P_L = 8.89\ W}$$

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Example 2



$$R_T = r_{source} + R_{load}$$

$$R_T = 10\ \Omega + 10\ \Omega$$

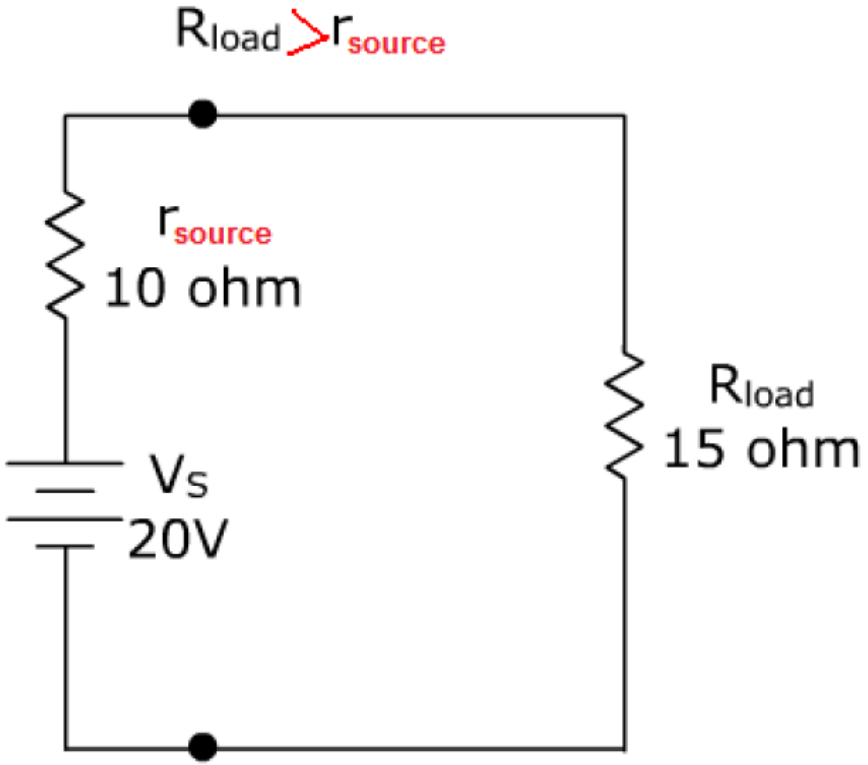
$$R_T = 20\ \Omega$$

$$I_T = \frac{V_S}{R_T} = \frac{20V}{20\Omega} = 1.00\ A$$

$$P_L = I^2 R_{load} = (1.00A)^2\ (10\Omega)$$

$$\mathbf{P_L = 10\ W}$$

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$R_T = r_{source} + R_{load}$

$R_T = 10\ \Omega + 15\ \Omega$

$R_T = 25\ \Omega$

$$I_T = \frac{V_S}{R_T} = \frac{20V}{25\Omega} = 0.80\ A$$

$$P_L = I^2 R_{load} = (0.80A)^2\ (15\Omega)$$

$P_L = 9.6\ W$

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Example 4.13

Find the value of R_L for maximum power transfer in the circuit of Fig. 4.50. Find the maximum power.

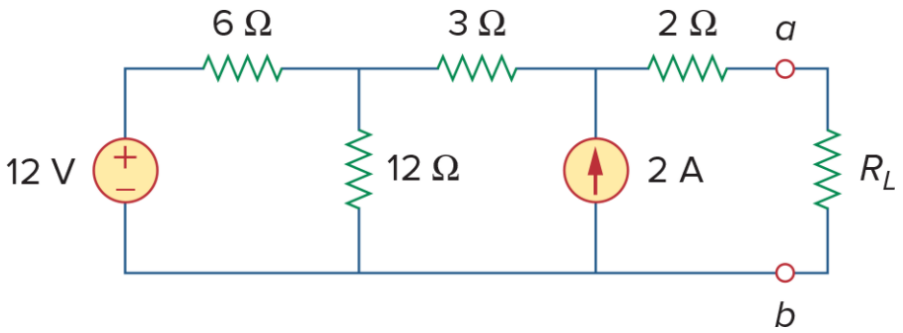


Figure 4.50
For Example 4.13.

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